NOTE

AWS Lambda is a serverless computing service offered by AWS that allows you to run code without provisioning or managing servers. You provide code to run and Lambda executes your code in response to events, managing all underlying compute resources, including server and operating system maintenance, scaling, and capacity provisioning. Microsoft's Azure functions and Google Cloud's cloud functions provide similar serverless compute functionality.

Figure 3.9 shows the architecture of the sensor data collection hub, where each sensor publishes data to an MQTT topic, and a single message decoder function subscribes to all sensor data topics, and is able to decode their payloads.

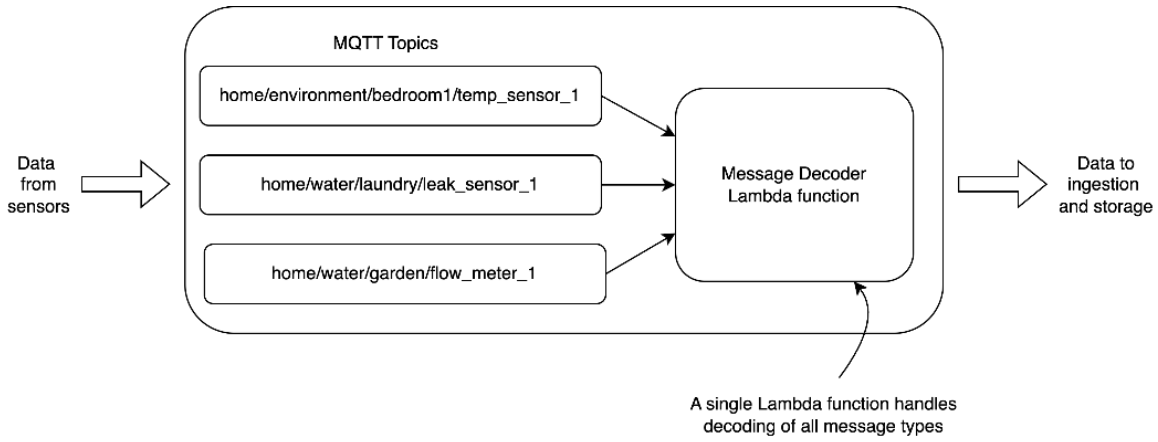


Figure 3.9 The sensor data collection hub architecture where messages from multiple types of sensor are collected, decoded, and routed for storage.

EXAMPLE: DECODING LORAWAN PAYLOADS

My message decoder Lambda subscribes to all MQTT topics that data is published to by sensors, and implements a decoding function for each type of message received. Consider my LHT52 indoor temperature and humidity sensors distributed around the house. When the one in the master bedroom transmits data, the following Base64 encoded hex string is received on the MQTT topic named `home/environment/bedroom1/temp_sensor_1` in AWS: